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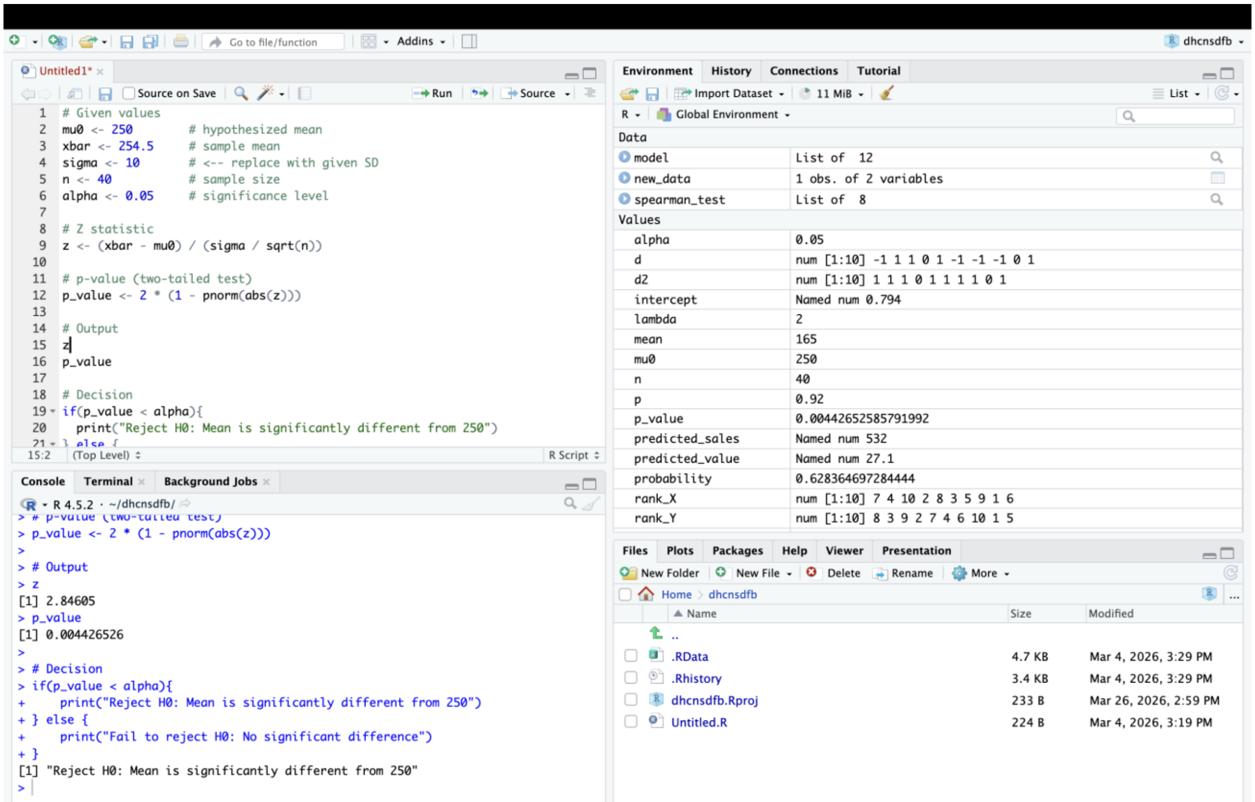
Probability and Statistics Lab (DA-4)

1. A pharmaceutical manufacturer produces tablets that are supposed to contain an average of 250 mg of an active ingredient. The process is known to have a population standard deviation of mg. To ensure the machines are calibrated correctly, a quality control engineer selects a random sample of 40 tablets and finds the mean content to be 254.5 mg. At the level of significance, is there sufficient evidence to conclude that the mean amount of the active ingredient is different from the claimed 250 mg?

CODE:



Que 1 (code part):-

Given values

mu0 <- 250 # hypothesized mean xbar <- 254.5 # sample mean sigma <- 10 # <-- replace with given SD n <- 40 # sample size alpha <- 0.05 # significance level

Z statistic

z <- (xbar - mu0) / (sigma / sqrt(n))

p-value (two-tailed test)

p_value <- 2 * (1 - pnorm(abs(z)))

Output

z p_value

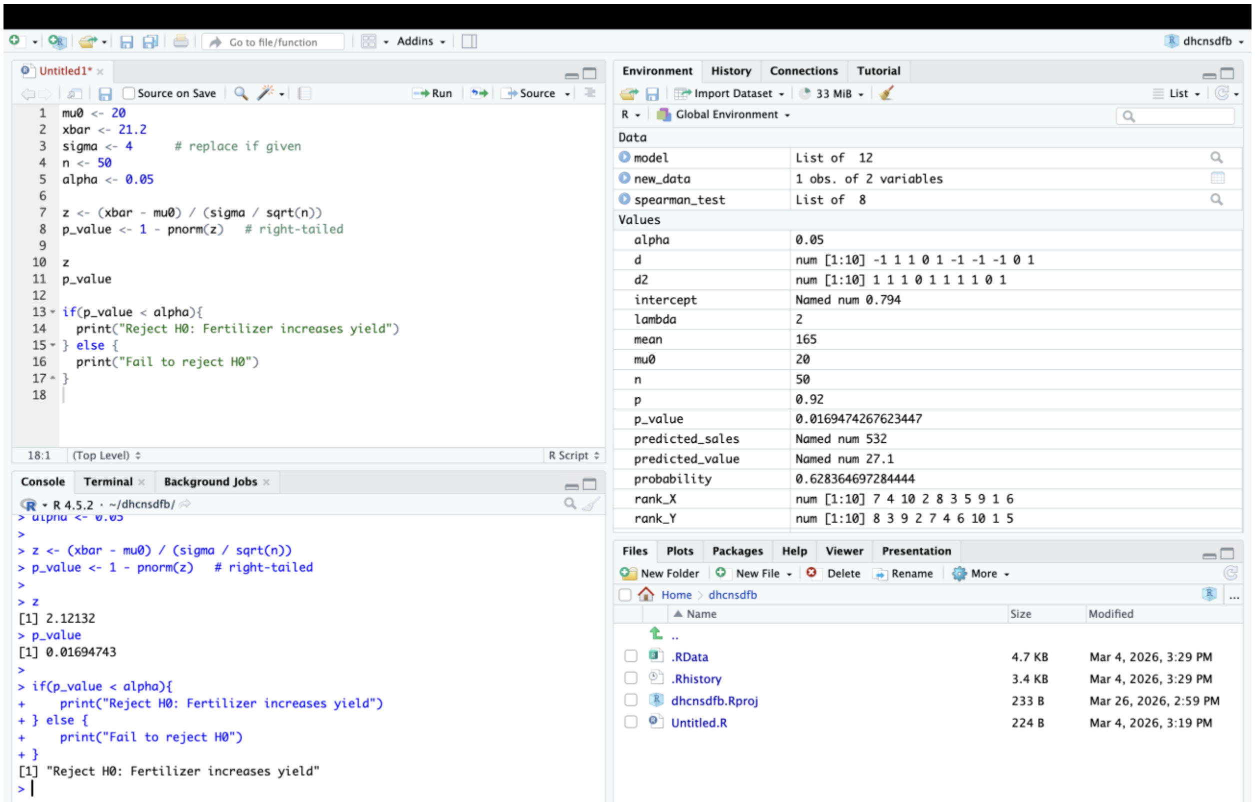
Decision

if(p_value < alpha){ print("Reject H0: Mean is significantly different from 250") } else
{ print("Fail to reject H0: No significant difference") }

QUE-2

1. A new organic fertilizer is claimed to increase the yield of tomato plants. Without the fertilizer, the national average yield is 20 lbs per plant with a known population standard deviation of lbs. A researcher applies the new fertilizer to a random sample of 50 plants and records an average yield of 21.2 lbs per plant. At the level of significance, does the data support the claim that the new fertilizer significantly increases the average yield?

CODE:-



Que –2(code part):-

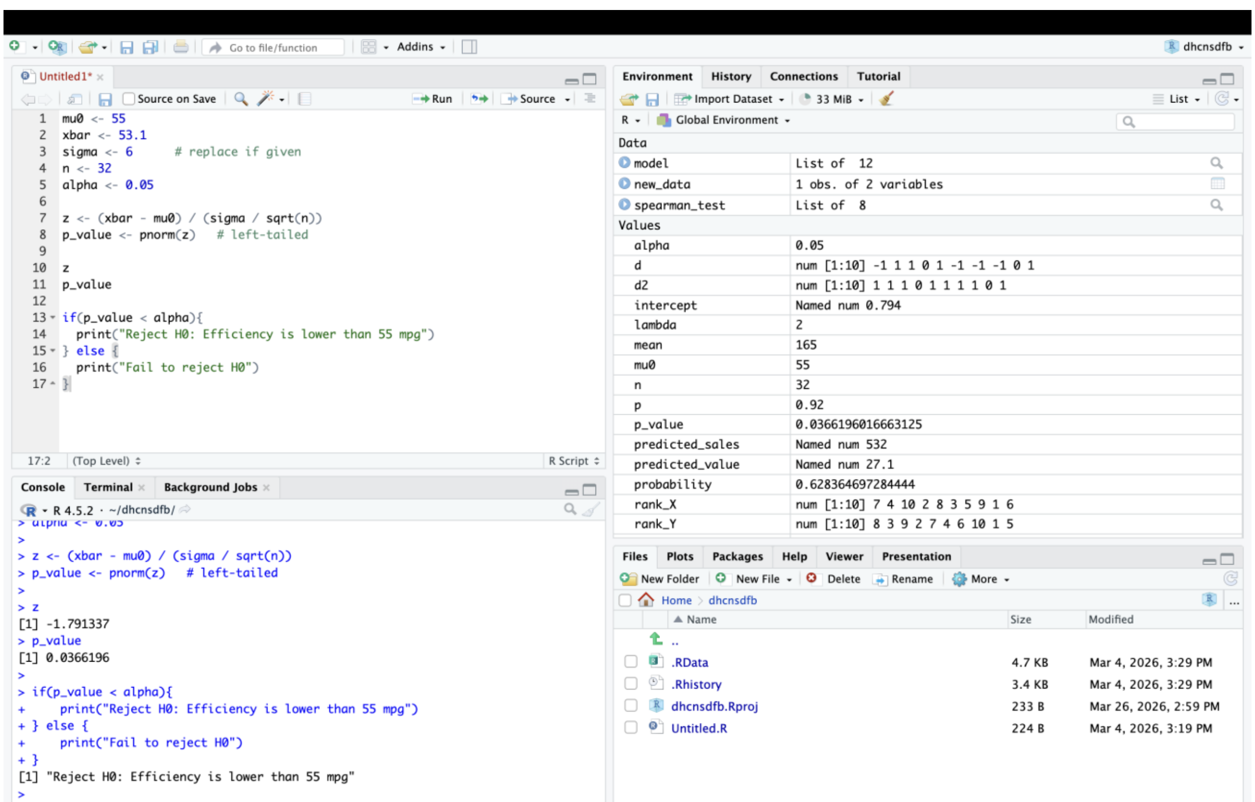
```
mu0 <- 20
xbar <- 21.2
sigma <- 4      # replace if given
n <- 50
alpha <- 0.05

z <- (xbar - mu0) / (sigma / sqrt(n))
p_value <- 1 - pnorm(z)   # right-tailed

z
p_value

if(p_value < alpha){
```

```
  print("Reject H0: Fertilizer increases yield")
} else {
  print("Fail to reject H0")
}
```

Que-3:-

1. A car manufacturer advertises that their new hybrid model achieves a mean fuel efficiency of 55 miles per gallon (mpg) on the highway. A consumer advocacy group suspects that the actual efficiency is lower than advertised. They know from independent testing that the population standard deviation is mpg. They test a random sample of 32 cars and find a sample mean of 53.1 mpg. Test the group's suspicion at the level of significance.

CODE:-



Que-3(code part):-

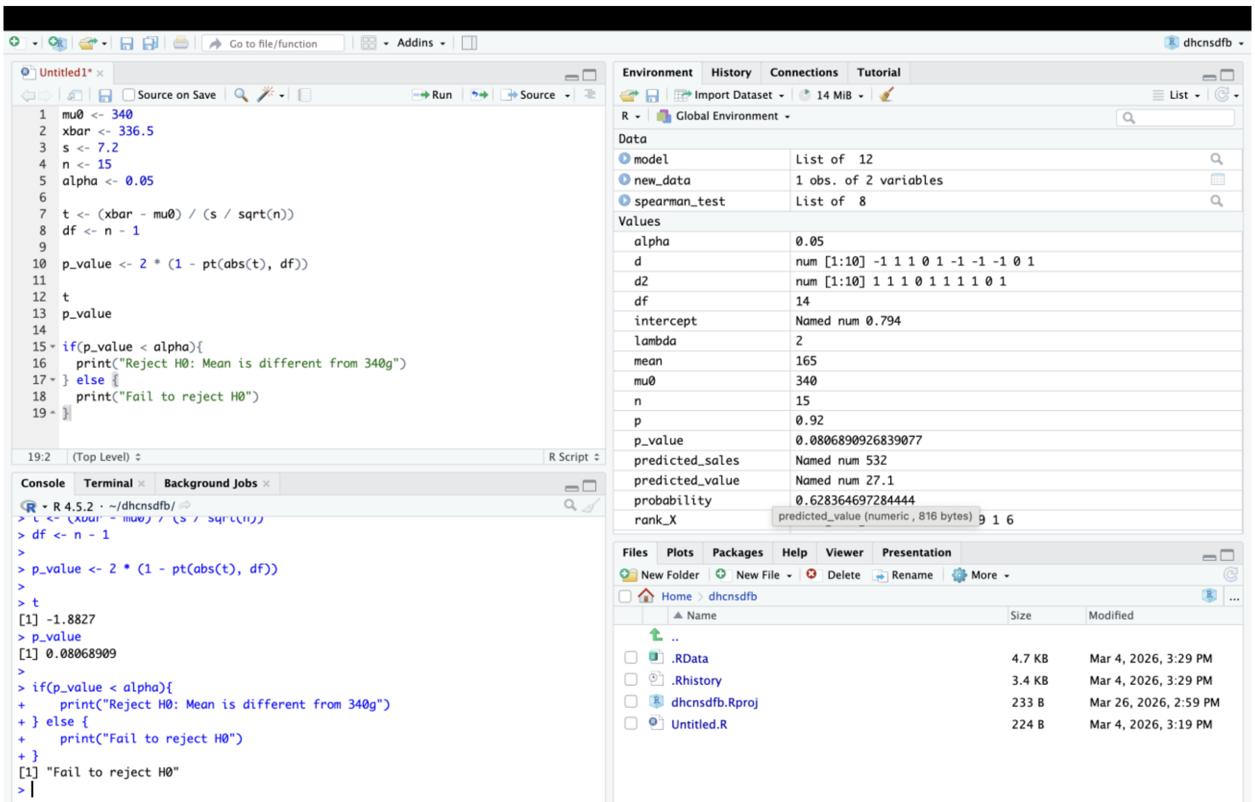
```
mu0 <- 55
xbar <- 53.1
sigma <- 6     # replace if given
n <- 32
alpha <- 0.05

z <- (xbar - mu0) / (sigma / sqrt(n))
p_value <- pnorm(z)   # left-tailed

z
p_value

if(p_value < alpha){
  print("Reject H0: Efficiency is lower than 55 mpg")
} else {
  print("Fail to reject H0")
}
```

Que-4:-

1. A boutique coffee roaster claims that their "Signature Blend" bags contain an average of 340 g of coffee beans. A consumer advocate suspects the bags are not filled accurately and might be over or under the limit. They collect a random sample of 15 bags and find a sample

mean of 336.5 g with a sample standard deviation of 7.2 g. At the 5% significance level ( ), is there sufficient evidence to conclude that the actual mean weight is different from the claimed 340 g?

Code:-



Question-4(code):-

mu0 <- 340

xbar <- 336.5

s <- 7.2

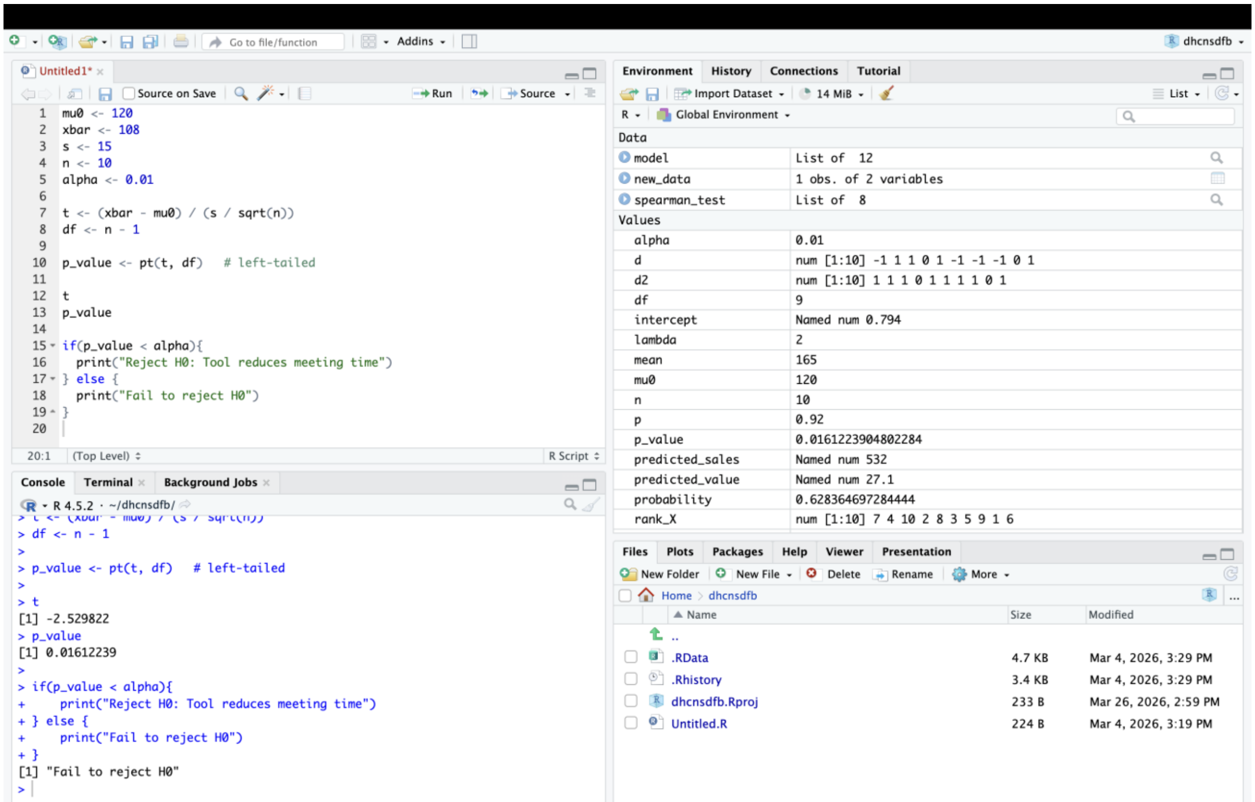
n <- 15

```
alpha <- 0.05


t <- (xbar - mu0) / (s / sqrt(n))

df <- n - 1


p_value <- 2 * (1 - pt(abs(t), df))


t

p_value


if(p_value < alpha){

  print("Reject H0: Mean is different from 340g")

} else {

  print("Fail to reject H0")

}
```

Que-5:-

1. A software company claims that its new project management tool reduces the average time spent on weekly meetings, which was previously 120 minutes per team. A consultant believes the tool is effective and tests it with a random sample of 10 teams. After one month, the teams report a mean meeting time of 108 minutes with a sample standard deviation of 15 minutes. At the 1% significance level ( ), does the data support the claim that the tool significantly reduces the average meeting time?

Code:-



Code-part:-

```
mu0 <- 120
xbar <- 108
s <- 15
n <- 10
alpha <- 0.01

t <- (xbar - mu0) / (s / sqrt(n))
df <- n - 1

p_value <- pt(t, df)   # left-tailed

t
p_value
```

```r
if(p_value < alpha){
  print("Reject H0: Tool reduces meeting time")
} else {
  print("Fail to reject H0")
}
```